

Benchmark notes for `lebedev_em`

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Abstract

These notes document the benchmarks for the `lebedev_em` solver, a Python implementation of the Lebedev staggered-grid finite-difference scheme of Davydycheva, Druskin & Habashy [1] (DDH03) for the frequency-domain Maxwell equations in 3-D anisotropic inhomogeneous media. Two inhomogeneous benchmarks are presented. The first, a two-half-space model with a $10\times$ conductivity contrast, is validated against the exact Sommerfeld-integral solution: the four-cluster Lebedev average attains $\approx 1\%$ accuracy in the conductive half-space, and cluster averaging reduces the single-cluster error fivefold. The second reproduces the configuration of DDH03 Figure 9 — a resistive borehole with an invaded zone crossing a thin 75° dipping anisotropic layer with $\sigma_N = \sigma_T/200$ — and matches the published curves at the few-percent level with unit dipole moment and no fitted parameters, *identically at every grid resolution*, using either the DDH03 eq.-(8/9) laminate (Backus) averaging or the nodal homogenization of [2] in a fully coupled four-cluster solve. A three-level refinement study shows that the two averaged schemes agree with each other to $\approx 1\%$ and are resolution-independent (reproducing DDH03’s own $k = 6$ vs. $k = 12$ insensitivity), whereas naïve pointwise sampling of the conductivity is off by 20% or more at every practical resolution and converges only slowly. The study also identifies a subtlety of the published model that an earlier version of these notes got wrong: the invasion annulus ($R = 0.6$ m), although it carries the background conductivity, *replaces the anisotropic layer* near the borehole, so the layer is present only at $r \geq 0.6$ m.

1 The solver and its verification chain

`lebedev_em` discretizes the frequency-domain Maxwell system (time convention $e^{-i\omega t}$)

$$\nabla \times \mathbf{E} = i\omega\mu\mathbf{H}, \quad \nabla \times \mathbf{H} = \dot{\sigma} \mathbf{E} + \mathbf{J}, \quad \dot{\sigma} = \sigma - i\omega\varepsilon, \quad (1)$$

on Lebedev’s staggered grid, which supports full 3×3 conductivity tensors without interpolation and splits into four Yee-like clusters. The implementation follows the complete DDH03 methodology: distributed sources in all four clusters satisfying their eq. (7) (unit total weight, centroid at the physical source), mixed electric/magnetic boundary conditions assigned per cluster so that domain-truncation errors cancel in the four-cluster average, interpolate-then-average field extraction, optimal geometric transverse grids ($\alpha = e^{\gamma\pi/\sqrt{k}}$, $\gamma = 1/\sqrt{2}$) with a hybrid axial grid (fine equidistant near source and receivers, geometric beyond), and sub-cell homogenization of interface-cut cells.

The verification chain proceeds from analytic anchors outward: the discrete operators are unit-tested (symmetry of the curl-curl system in its volume-weighted inner product, source weight and centroid conditions on nonuniform grids); the solver reproduces the closed-form magnetic-dipole

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solution in homogeneous isotropic media at both 2.5 kHz and 52.65 kHz to a few percent at moderate resolution; and the two inhomogeneous benchmarks below add a sharp material interface and a thin anisotropic structure. All results are per unit dipole moment ($1 \text{ A}\cdot\text{m}^2$) in SI units, with no calibration constants anywhere in the chain.

2 Benchmark 1: two half-spaces, $10\times$ contrast

Model. A unit vertical electric dipole (VED) at the origin in

$$\sigma(z) = \begin{cases} 0.1 \text{ S/m}, & z < 4 \text{ m}, \\ 1.0 \text{ S/m}, & z \geq 4 \text{ m}, \end{cases} \quad f = 2500 \text{ Hz},$$

with receivers on the z -axis, $z \in [0.5, 7.5] \text{ m}$, straddling the contact. The reference is the exact Sommerfeld-integral solution for E_z (implemented in `analytics.py` and independently verified: it reduces to the closed-form homogeneous solution when $\sigma_1 = \sigma_2$ to 3×10^{-14} relative, and satisfies the normal-current continuity $\sigma_1 E_z^- = \sigma_2 E_z^+$ at the contact).

Grid. Optimal geometric transverse grids with $k = 3$ ($M_x = M_y = 12$), hybrid axial grid with $\Delta z = 0.0625 \text{ m}$ near the source ($M_z = 160$, $\approx 41\text{k}$ complex degrees of freedom).

Results. Figure 1 shows $|E_z|$ on the axis, the pointwise deviation from the analytic solution, and the contact effect.

- In the conductive half-space ($z \geq 4 \text{ m}$), across the $10\times$ jump, the RMS relative error of $\text{Re } E_z$ is **1.0%**.
- The overall RMS error (5.6%) is dominated by the near-source region, where the transverse domain is only $0.3 \delta_1$ wide; the *same* error appears in a homogeneous control run (RMS 5.7%), so the interface itself contributes essentially nothing — the scheme crosses the contact cleanly.
- The physical structure at the contact is captured quantitatively: the two-layer/homogeneous ratio rises to 1.65 just above the contact and drops to ≈ 0.18 below it, matching the analytic curve point by point.
- Cluster averaging matters: a single cluster (C000) gives RMS 29.9%; the four-cluster Lebedev average reduces this to 5.7% — DDH03’s error-cancellation mechanism at work in an inhomogeneous medium.

3 Benchmark 2: thin dipping anisotropic layer crossed by a borehole

Model. The configuration of DDH03 Figure 9 (Fig. 2): a homogeneous background $\sigma = 0.1 \text{ S/m}$; a resistive borehole column ($\sigma = 0.05 \text{ S/m}$, $R = 0.1 \text{ m}$) along the z -axis; an invaded zone ($\sigma = 0.1 \text{ S/m}$, $R = 0.6 \text{ m}$); and a thin dipping anisotropic layer of 0.25 m normal thickness, dip 75° , with transverse conductivity $\sigma_T = 0.1 \text{ S/m}$ and normal conductivity $\sigma_N = \sigma_T/200$ — a 1:200 contrast confined to a layer thinner than most local grid cells. The layer crosses the borehole axis over $z \in [-1.93, -0.95] \text{ m}$. The source is a z -directed unit magnetic dipole at the origin, $f = 52.65$

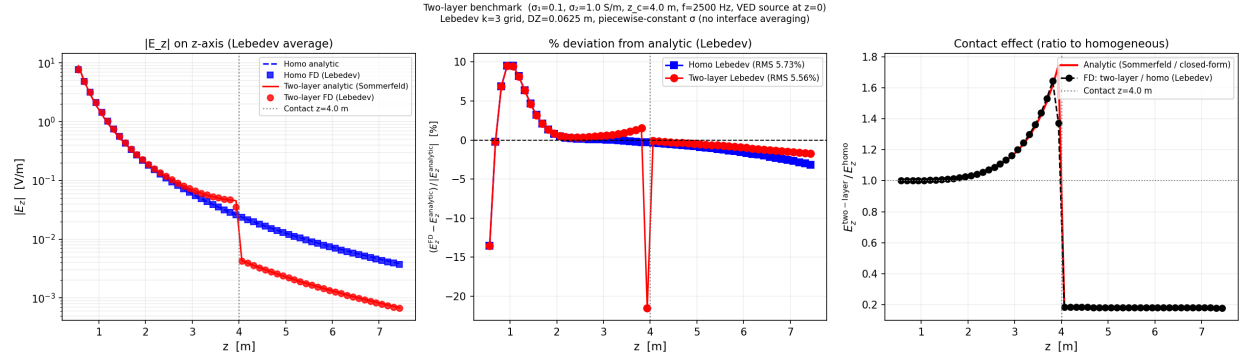


Figure 1: Two-half-space benchmark ($\sigma_1 = 0.1$, $\sigma_2 = 1.0$ S/m, contact at $z_c = 4$ m, $f = 2500$ Hz). Left: $|E_z|$ on the axis, FD (symbols) vs. analytic (lines). Center: pointwise percent deviation. Right: contact effect — two-layer to homogeneous ratio, FD (black) vs. Sommerfeld analytic (red). The isolated spike in the center panel sits at the contact node itself, where E_z is discontinuous and a pointwise comparison is ill-posed.

kHz; the observable is $\text{Im } B_z$ on the axis. Two cases are run: *no layer* (borehole only) and *resistive layer*.

A modelling subtlety, corrected in this revision, turned out to control the comparison. The invasion annulus carries the *same* conductivity as the background, and an earlier version of these notes therefore omitted it. But in the DDH03 model family (cf. their Figs. 6–7, where the invasion visibly punches through the layered formation) the invaded zone *replaces* the formation near the wellbore: the anisotropic layer exists only for $r \geq 0.6$ m. Because the induction tool is most sensitive precisely in that near-axis region, the two readings differ enormously — modelling the layer all the way in to the borehole wall increases the layer’s attenuation of $\text{Im } B_z$ from $\approx 27\%$ to $\approx 51\%$ (geometric mean over the receivers) and cannot be reconciled with the published curve by any choice of scheme or resolution.

Method. Full DDH03 methodology as in §1: optimal transverse grids, hybrid axial grid with the source exactly on a grid node, four-cluster sources and mixed boundary conditions, exact-geometry sub-cell homogenization (exact interface normals and volume fractions from the analytic geometry) by either the DDH03 eq.-(8/9) laminate (“Backus”) scheme or the nodal scheme of [2], a *single fully coupled* sparse iterative solve (the off-diagonal conductivity couples the four clusters; solving them separately under-counts the anisotropy-generated cross-components), and interpolated four-cluster averaging at the exact receiver positions. Three refinement levels are run, scaling the transverse minimal step and the axial spacing together while growing k to hold the domain reach: L1 ($h_{\min} = 0.05$ m, $k = 6$, $\approx 121k$ unknowns — the published-comparison grid), L2 ($h_{\min} = 1/30$ m, $k = 7$, $\approx 223k$), and L3 ($h_{\min} = 0.025$ m, $k = 8$, $\approx 368k$). All iterative solves converged to 10^{-8} relative residual.

Absolute calibration of the reference. The published “no layer” curve serves double duty: comparing it to the closed-form homogeneous-medium solution shows agreement within 1–8% (the deficit appearing near the source, consistent with the borehole’s local reduction of the field). This confirms that the published figure is in absolute SI units per unit dipole moment, so the comparison below involves no free scale.

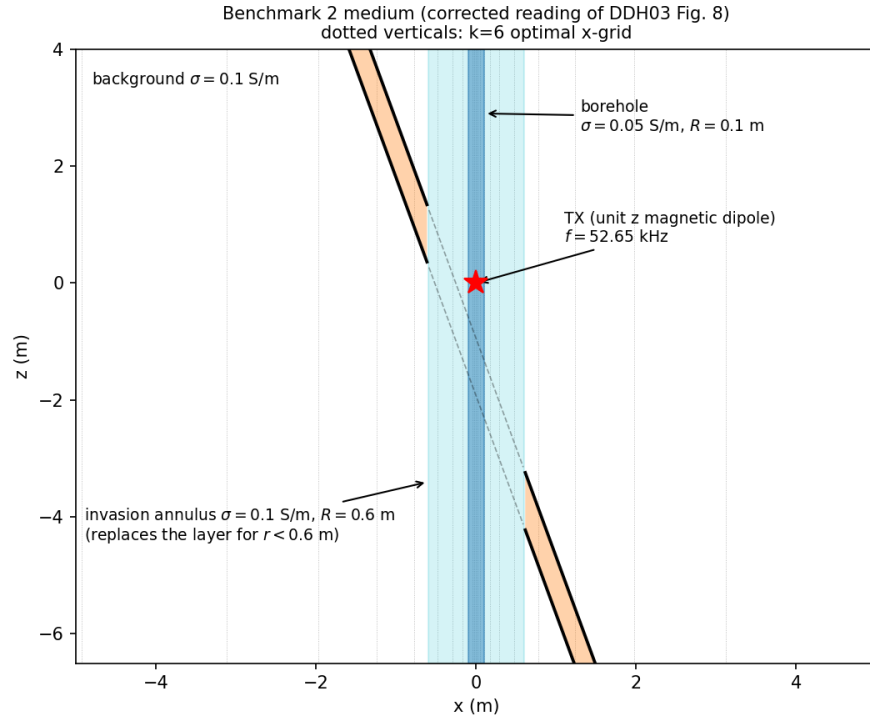


Figure 2: Benchmark 2 medium, x - z section (corrected reading of DDH03 Fig. 8). The invasion annulus ($\sigma = 0.1$ S/m, $R = 0.6$ m) coincides with the background conductivity but replaces the anisotropic layer for $r < 0.6$ m, so the layer (orange) is an annular structure. Dotted verticals show the $k = 6$ optimal transverse grid.

Results. Figure 3 overlays the computed curves at the finest grid (lines) on the values digitized from the published figure (symbols).

- **No layer:** agreement within 0.3–4.5% at every receiver (geometric-mean ratio 1.021 Backus, 1.019 nodal at L3; log-scatter 1.1%), improving steadily with refinement (1.035 at L1). This confirms the absolute normalization and sets the accuracy floor of the digitization itself at the 1–2% level.
- **Resistive layer (invaded model):** agreement within 3–12% (geometric-mean ratio 1.070 Backus, 1.077 nodal at L3; log-scatter 2%), and — crucially — *flat across all three refinement levels*: the Backus ratio moves only $1.085 \rightarrow 1.073 \rightarrow 1.070$ from L1 to L3, and Backus and nodal agree with each other to better than 1.2% at every receiver. In the grid-calibration-free attenuation measure (geometric mean of the layer/no-layer ratio over the receivers) the published figure gives 0.697 and the averaged schemes give 0.729–0.742 at every level. The residual few percent is consistent with digitization error: both the layer’s axis intercepts ($-0.95/-1.93$ m) and the published values themselves are read off the printed figure.

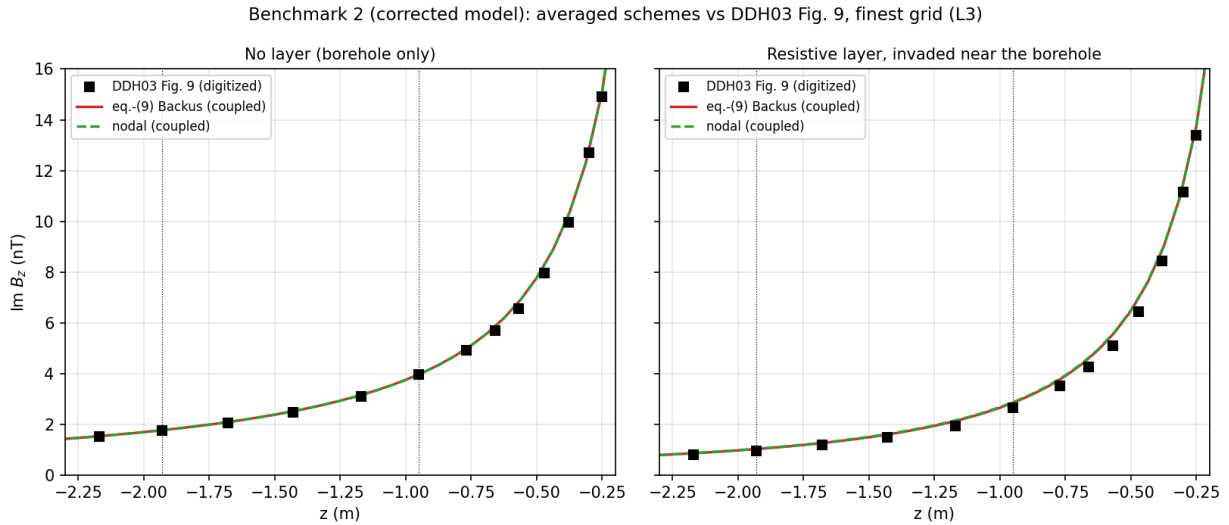


Figure 3: Benchmark 2: $\text{Im } B_z$ on the axis at the finest grid (L3). Lines: `lebedev_em`, coupled solve with eq.-(9) Backus (solid red) and nodal (dashed green) averaging — the two are indistinguishable at this scale. Symbols: values digitized from DDH03 Fig. 9. Unit dipole moment on both sides; no fitted constants. Dotted verticals mark where the layer crosses the axis.

Refinement study: averaging is required, and the model matters more than the scheme.

Figure 4 summarizes the three-level refinement study for both readings of the model and all three sub-cell treatments. Three conclusions:

1. **The two averaged schemes are resolution-independent and equivalent.** On the correct (invaded) model, Backus and nodal sit within $\approx 7\%$ of the published curve at every level and within $\approx 1\%$ of each other — reproducing DDH03’s own observation that their $k = 6$ and $k = 12$ curves “can hardly be distinguished,” a property that only holds for averaging schemes. This is the frequency-domain counterpart of the thin-resistive-structure results of [2]: the sub-cell layer is captured by homogenization, not by resolution.

2. **Pointwise sampling is inadequate at any practical resolution.** On the invaded model it overshoots the published curve by 26% at L1 and is still 22% high at L2; in a controlled tilted-slab study with the same layer material it remains 23% high even at $t/h = 2.6$ and converges only slowly from above.
3. **The earlier contrary conclusion was a model artifact.** On the *uninvaded* model (layer extending to the borehole wall), pointwise at the published-comparison grid happens to land on the digitized curve (attenuation 0.698 vs. the paper’s 0.697) — a double coincidence in which pointwise’s failure to see the sub-cell layer imitates the genuinely absent near-axis layer of the true model. Refinement unmasks it: the pointwise curve drifts steadily down through and past the published values ($1.036 \rightarrow 1.008 \rightarrow 0.982$ of the paper from L1 to L3, still falling), while the averaged schemes on the same wrong model sit 30% low at every level. No scheme reproduces the published curve on the uninvaded model; both averaged schemes reproduce it on the invaded one.

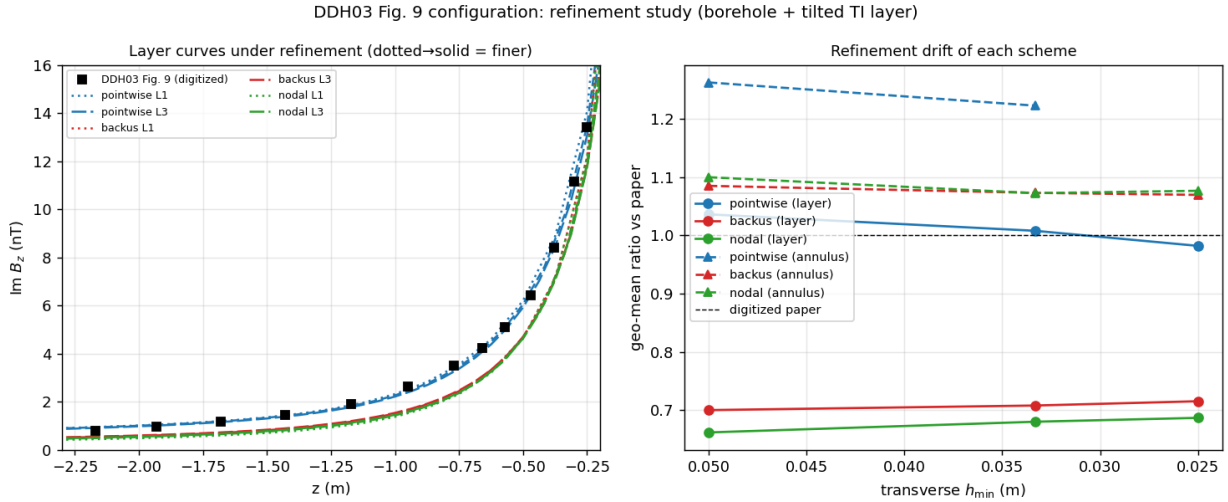


Figure 4: Three-level refinement study of Benchmark 2. Left: the resistive-layer curves for the uninvaded model under refinement. Right: geometric-mean ratio to the digitized published values for each scheme, on the uninvaded (“layer,” circles) and invaded (“annulus,” triangles) models. The averaged schemes on the invaded model (red/green triangles) are flat at ≈ 1.07 ; pointwise is resolution-dependent on either model.

4 Remarks

The two benchmarks are complementary: Benchmark 1 has an exact external reference and tests the sharp-interface machinery and cluster averaging; Benchmark 2 tests the full anisotropic pipeline — tilted tensors, a borehole crossing a thin high-contrast layer, homogenization, the coupled four-cluster solve — against independently published curves whose absolute normalization we verified analytically, and its refinement study additionally validates the resolution-independence that is the central claim of the DDH03 averaging approach. Both succeed with unit dipole moment and no adjustable constants.

Benchmark 2’s history is itself instructive. The earlier version of these notes reported that only the nodal scheme (run in a then-current sequential per-cluster mode) reproduced the published curve. Subsequent work replaced the sequential solves by the fully coupled solve, verified the eq.-(9) averaging, the coupled off-diagonal operator, and both averaged schemes against analytic and refinement ground truths in a chain of controlled tests (tilted whole-space, single tilted interface, tilted sub-cell slab), and traced every remaining discrepancy with the published figure to the model-transcription error described in §3: the omitted invaded zone. The lesson we would pass on: when a benchmark against a digitized figure disagrees, refine the grid before trusting the match — the scheme that agreed at the published resolution was the one that was wrong.

Work in progress: the deviated-borehole model of DDH03 Figs. 6–7 (borehole, invasion zone, and a 60° dipping anisotropic half-space) reproduces the published curve *shapes* precisely but not yet their amplitudes. The discrepancy had independently been localized to the treatment of the invasion zone in that figure; in light of the Benchmark-2 resolution — the invaded zone replaces the formation, including any anisotropic layers it crosses — that model is being revisited and is omitted from these notes until it is settled.

References

- [1] S. Davydycheva, V. Druskin, T. Habashy, *An efficient finite-difference scheme for electromagnetic logging in 3D anisotropic inhomogeneous media*, Geophysics **68**(5):1525–1536, 2003.
- [2] S. Moskow, V. Druskin, T. Habashy, P. Lee, S. Davydycheva, *A finite difference scheme for elliptic equations with rough coefficients using a Cartesian grid nonconforming to interfaces*, SIAM J. Numer. Anal. **36**:442–464, 1999.